

Khushei Meghana Meda

Amherst, MA | [Linkedin](#) | [Github](#) | [Portfolio](#) | +1 (413) 315-1495 | kmeda@umass.edu

EDUCATION

University of Massachusetts Amherst

Amherst, USA

Master of Science in Computer Science (GPA: 3.95/4)

Sep 2025 - May 2027 (Expected)

- Coursework: Robotics, Advanced Natural Language Processing, Machine Learning, Data Science

PES University

Bengaluru, India

Bachelor of Technology in Computer Science and Engineering (GPA: 9.04/10)

Aug 2018 - May 2022

- Awards: M R Doreswamy Scholarships, given to the top 20% of the batch of 600 CSE students for four semesters
- Research: Published "Teaching Aid For Program Optimization" in IEEE Xplore and presented at IEEE 2021 SMARTGENCON

TECHNICAL SKILLS

Languages: Java, Python, Javascript, Typescript, C++, HTML, CSS/SCSS, C, SQL

Technologies/Frameworks: Spring Boot, ReactJs, Redux, Node.js, Django, Docker, Iceberg, Prometheus, Grafana, Splunk, ROS, gRPC

Databases/Cloud: PostgreSQL, MongoDB (NoSQL), Redis, Kubernetes, GCP, AWS

Other: Git, Linux, Claude Code, Cursor, Bazel, Jenkins, CI/CD, Protobuf, microservices, UI/UX, OOP, MCP, Agents, LLMs, SDLC

EXPERIENCE

Zoox

Foster City, USA

Software Engineering Intern | C++, Typescript, ROS

May 2026 – Aug 2026

- Scaled and optimized the vehicle-to-cloud communication middleware at Zoox, supporting real-time data flow between the autonomous robotaxi fleet and downstream web services including teleoperation and fleet management systems.
- Migrated real-time video streams from WebSockets to WebTransport over QUIC to reduce latency and staleness, enabling more responsive consumption by fleet operations and teleoperation services under bad network conditions.
- Added observability to measure live stream video quality metrics including latency, jitter, and staleness across vehicle sessions to monitor and validate streaming performance.

Center for Data Science and AI, University of Massachusetts Amherst

Amherst, USA

Full-Stack Software Engineer | Typescript, Javascript

Nov 2025 – May 2026

- Created a university-wide GenAI platform with 4000+ users by extending LibreChat, an open-source unified orchestration layer for multiple LLM providers, models, and agents.

Apple

Bengaluru, India

Software Engineer | Java, Javascript

Jul 2022 – Aug 2025

- Architected and implemented scalable data (ETL) pipelines processing 2-3TB time series telemetry data daily from 2000+ hosts into a centralized data lake, and created REST APIs and custom UI interfaces for analytics platform.
- Built the core worker module for queuing, distribution, and scheduling requests across 350+ Kafka clusters with 1500+ nodes deployed on cloud infrastructure.
- Led design and development of the Kafka-as-a-Service self-service portal, building REST APIs, database schemas, and interactive UI components that reduced operational overhead by 70%.
- Designed a chatbot with agentic workflows for the KaaS portal, saving admins 30 minutes daily with intelligent query handling, and mentored an intern on its implementation.
- Developed custom web dashboards to visualize Teradata infrastructure metrics and KPIs, enabling data-driven capacity planning and disaster recovery operations.

VMware

Bengaluru, India

Software Engineering Intern | Java

Jan 2022 - Jun 2022

- Improved resilience of a cluster autoscaling service by developing a bot to assist in handling customer incidents, monitor production pipelines, report bugs, and automate developers' daily tasks.
- Automated incident triaging and bug reporting which improved process time by 10x from 5 minutes to 30 seconds.

PROJECTS

Multi-Goal Delivery Robot with Dynamic Replanning | ROS1, Gazebo, Rviz, Python

[Github](#)

- Implemented an end-to-end ROS1 delivery stack on an omnidirectional Triton robot that maps an indoor space, plans paths with A*, and visits multiple drop-off locations in an efficient order.
- Applied LiDAR-based obstacle checks and on-the-fly replanning so the robot stops and reroutes when its path is blocked, then continues the remaining deliveries.

Autonomous Robot Wall Following | ROS1, Gazebo, Python

[Github](#)

- Implemented Q-Learning and SARSA in ROS1 with custom state representations from 2D LiDAR sensor data and reward functions to optimize Triton robot navigation in Gazebo simulation.
- Applied epsilon-greedy exploration with Temporal Difference learning to iteratively update Q-values, enabling the robot to generalize the learned policy across five distinct wall-following scenarios.

Robot Localization | ROS1, Gazebo, Rviz, Python

[Github](#)

- Implemented a particle filter (Monte Carlo Localization) in Python and ROS to localize a LiDAR-equipped robot in a simulated environment using an odometry-based motion model, likelihood field sensor model, and low-variance resampling.